

Università
degli Studi
di Padova

Messa a punto di un potenziale interatomico per la simulazione della tantala amorfa tramite machine-learning

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quantum espresso



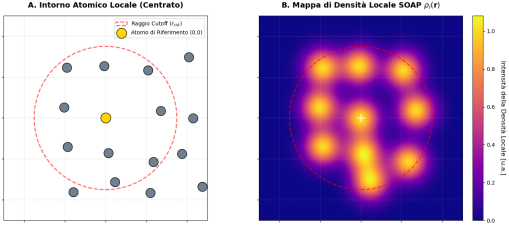


SOAP

SOAP: Smooth Overlap of Atomic Positions

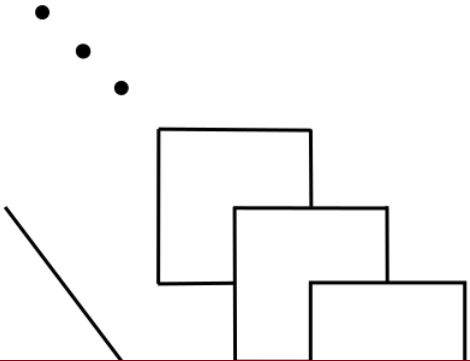
Generazione della mappa di densità

$$\rho_Z(\vec{r}) = \sum_{\alpha}^{Z} e^{-\frac{|\vec{r}-\vec{r}_\alpha|^2}{2\sigma^2}} \qquad (1)$$



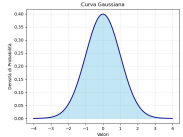
SOAP

dinamica molecolare suddivisa in fra

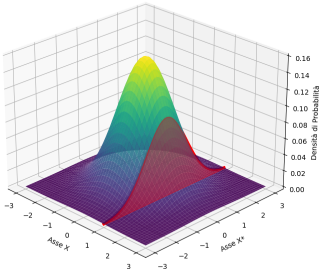


GAP





Gaussiana bivariata



Simulazione NVE

Simulazione dove si impone la conservazione di:

- numero di particelle
- volume
- energia

ZBL: Potenziale Ziegler-Biersack-Littmark

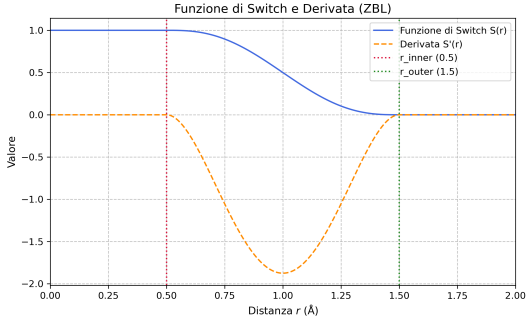
$$V(r_{ij}) = \frac{1}{4\pi\epsilon_0}\frac{Z_i Z_j}{r_{ij}}\phi\left(\frac{r_{ij}}{a_{ij}}\right) \tag{6}$$

$$\phi\left(\frac{r_{ij}}{a_{ij}}\right) = \phi(x) = 0.1818e^{-3.2x} + 0.5099e^{-0.9423x} + \\ + 0.2802e^{-0.4029x} + 0.02817e^{-0.2016x} \tag{7}$$

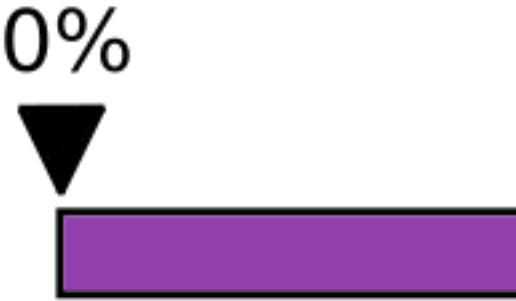
$$a_{ij} = \frac{0.8854\, a_0}{Z_i^{0.23} + Z_j^{0.23}} \tag{8}$$

$$S(u) = -6u^5 + 15u^4 - 10u^3 + 1 \tag{9}$$

$$u = \frac{r - r_{inner}}{r_{outer} - r_{inner}} \tag{10}$$



Simulazione con algoritmo di Metropolis



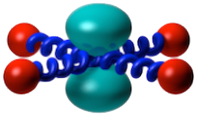
Python

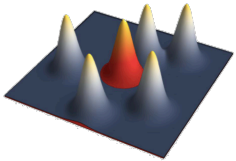
L'analisi è stata svolta in Python utilizzando le seguenti librerie:

- ASE: gestione computazionale della struttura fisica
- DScribe: test del metodo SOAP sui dati
- Quippy: addestramento del potenziale





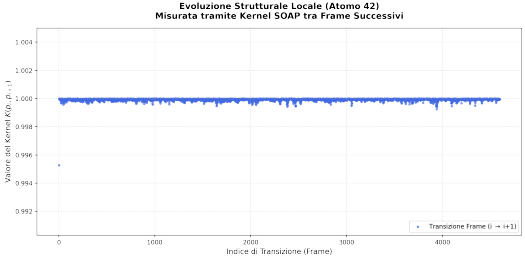


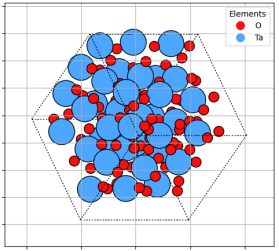


Dataset A

Divisione dei dati: Training set - Test set

Verifica dei risultati soap

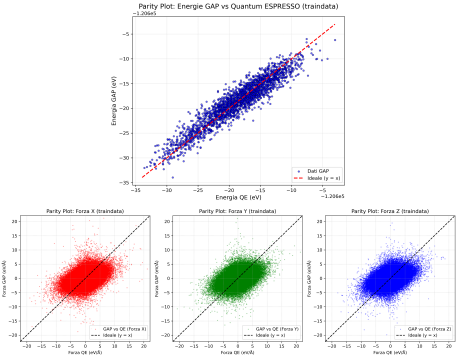




Frame del dataset A generato con matplotlib

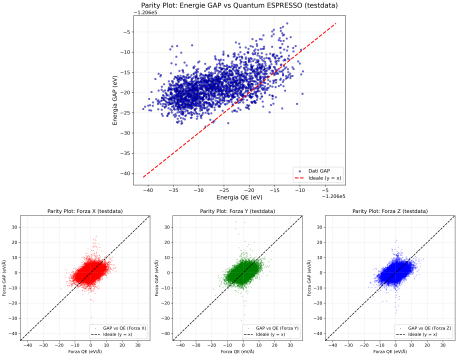
Verifica delle energie e delle forze

Training set



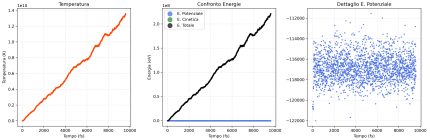
Energie e forze divise per componenti

Test set

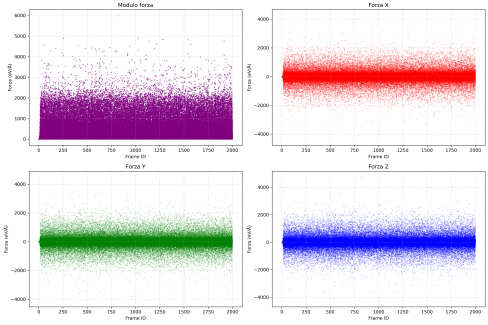


Energie e forze divise per componenti

Simulazione NVE



Andamento temporale di temperatura ed energia

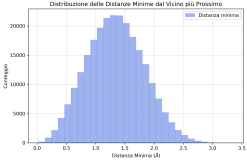


Analisi delle forze, divise per componenti e totale

Simulazione NVE: analisi

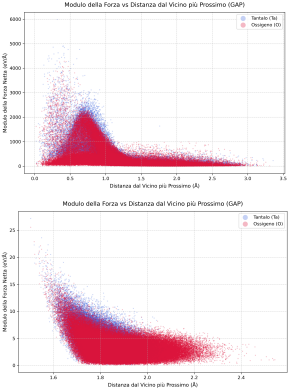
Distanza tra i primi vicini

Set di dati simulato con GAP



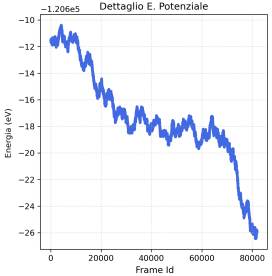
Set di dati di training

Relazione forza-distanza

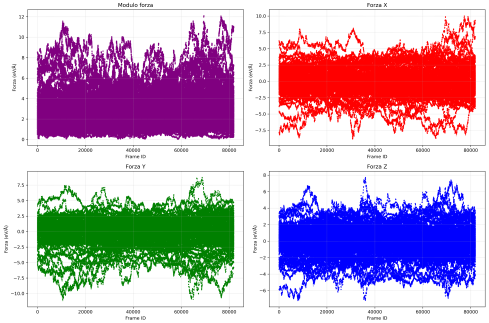


ZBL: risultati

Algoritmo di Metropolis: risultati

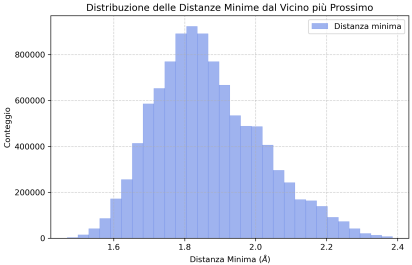


Andamento temporale dei valori di energia

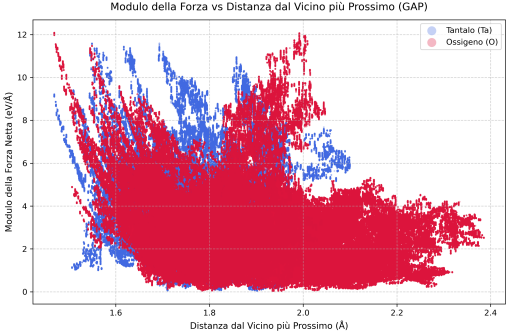


Analisi delle forze, per componenti e totale

Algoritmo di Metropolis: risultati



Distanza tra i primi vicini



Relazione forze-distanza